Rate amplification for non-adaptive linear locally correctable codes via dual smooth recovery

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Editorial note. Faithful editable transcription from the official CCC2021 CC BY4.0 published version, article1. One rate-amplification mechanism: introductory Theorem3 and formal Theorem46/Proposition55 with all of Sections4/4.1/4.2/4.3/4.4 author constructions and proofs, plus the full necessary Section5.2 axis-evasive partition construction and Claims57–63/Proposition60/Theorem62 proof. Definitions1–2/non-adaptive/linear code context and Claim82 retained. The unused probabilistic alternative Section5.1 is outside this item, as the author explicitly says it is not used; no separately counted support lemma. All43 original bibliography entries retained. Author wording/printed indices and equations are preserved without mathematical correction, including Definition25 distinct/not-necessarily-distinct wording, Claim23 z direction, Corollary49 final displayed little-o sign, Proposition60 final denominator c, Claim59 property numbering and final Theorem62 printed I indices. Author omitted details remain omitted rather than supplied: Fact24 called easy to verify, Claim82 induction, and Proposition55 final error calculation left to reader. These are the authors actual proofs, not extraction omissions. Formal54 parity wording retained as printed. All field/linearity/nonadaptivity/rate/error/smoothness/alphabet and prime-residue conditions retained. Root handles compilation/database/publication.

Allowed standard prerequisites (editorial).

- Siegel-Walfisz theorem. Original Theorem48[34,37] statement and full author Corollary49 deduction retained.
- Basic finite-field and cyclic-group facts. Full Section5.2 author use and supporting calculations retained; no recursion into standard algebra.

Theorem 46 (Author Theorem46)**.** *There exist universal constants $m_0, c' \geq 1$ such that the following holds. Let P be a set of size $m \geq m_0$. Let $\mathbb{F}$ be a field, and let $(D^P_{\text{in}}, L^P_{\text{in}})$ be a $(q_{\text{in}}, \tau_{\text{in}}, \varepsilon_{\text{in}}, \rho_{\text{in}})$-dual SLR over $\mathbb{F}^P$. Let $0 < \alpha < 1$ be such that*

$$\rho_{\text{in}} \geq \frac{c'}{\sqrt{\alpha \cdot \log m}} \log\left(\frac{1}{\alpha}\right). \tag{4.11}$$

Then, there exists a $(q_{\text{out}}, \tau_{\text{out}}, \varepsilon_{\text{out}}, \rho_{\text{out}})$*-dual SLR* $(D_{\text{out}}^{P_{\text{out}}}, L_{\text{out}}^{P_{\text{out}}})$ *over* $\mathbb{F}^{P_{\text{out}}}$*, with* $m^\ell/2 \leq |P_{\text{out}}| \leq m^\ell$*, where*

$$\ell = \Theta\left(\frac{1}{\rho_{\text{in}}} \log \frac{1}{\alpha}\right), \tag{4.12}$$

having the following parameters:

$$\begin{aligned}
q_{\text{out}} &\leq q_{\text{in}}^{\text{poly}(\ell)}, \\
\tau_{\text{out}} &\leq q_{\text{in}}^{\text{poly}(\ell)} \tau_{\text{in}}^{\ell}, \\
\varepsilon_{\text{out}} &\leq q_{\text{in}}^{\text{poly}(\ell)} (\tau_{\text{in}} + \varepsilon_{\text{in}}), \\
\rho_{\text{out}} &\geq 1 - \alpha.
\end{aligned}$$

Author code definitions and introductory main result

Definition 1 (Author Definition1: Locally decodable codes (LDC))**.** *A code* $C : \Sigma^k \to \Sigma^n$ *is* (q, δ, ε)*-locally decodable if there exists a randomized algorithm* D*, called a local decoder, that is given* $i \in [k]$ *as input and an oracle access to* $z \in \Sigma^n$*, and has the following guarantee. For every* $i \in [k]$*,* $m \in \Sigma^k$ *and* $z \in \Sigma^n$ *such that* $\text{dist}(C(m), z) \leq \delta$ *it holds that* $\Pr[D^z(i) \neq m_i] \leq \varepsilon$*. Moreover,* D *makes at most* q *queries to* z*.*

Definition 2 (Author Definition2: Locally correctable codes (LCC))**.** *A code* $C \subseteq \Sigma^n$ *is* (q, δ, ε)*-locally correctable if there exists a randomized algorithm* D*, called a local corrector, that is given* $j \in [n]$ *as input and an oracle access to* $z \in \Sigma^n$*, and has the following guarantee. For every* $j \in [n]$*,* $c \in C$ *and* $z \in \Sigma^n$ *such that* $\text{dist}(c, z) \leq \delta$ *it holds that* $\Pr[D^z(j) \neq c_j] \leq \varepsilon$*. Moreover,* D *makes at most* q *queries to* z*.*

We place z in the upper script in our notation $D^z(i)$ to stress that the number of symbols read from z by D is of importance. The parameter q is called the query complexity, and δ is the local error decoding radius, in the case LDC, and local error correction radius in the case of LCC. However, we also refer to δ, somewhat inaccurately, as the local distance of the code. From here on, we do not make any explicit reference to the "global" distance of a code and so we refer to the local distance simply as the distance. Throughout the paper, we only consider non-adaptive LDC and LCC, defined next. Informally, these are code in which the local decoder (or corrector) samples the entries to be read before the querying step takes place. Our results only hold for non-adaptive LDCs and LCCs. For ease of discussion, throughout the introduction we ignore the error parameter ε. More precisely, when stating our results, every LDC or LCC (both in the hypothesis as well as in the LDC or LCC guaranteed by the theorem) has constant error. Clearly, an LCC with a systematic encoding[1] is also an LDC and so, in particular, linear LCC induce LDC.

[1] An encoding from messages to codewords is called systematic if the symbols of each message are embedded in its mapped codeword.

Theorem 3 (Author introductory Theorem3: Main result)**.** *Assume one has a non-adaptive linear $(q, \delta = \Omega(1))$-LCC with block-length n_0 having rate $\rho = \rho(n_0)$. Then, for every integers $\ell, c \geq 1$ such that $\ell^2 < c < \log n_0$, one can obtain a non-adaptive linear $(q_{\text{new}}, \delta_{\text{new}})$-LCC with block length $n \approx n_0^{\ell}$, having rate ρ_{new}, where*

$$\begin{aligned} q_{\text{new}} &= (cq)^{\text{poly}(\ell)}, \\ \delta_{\text{new}} &= (cq)^{-\text{poly}(\ell)}, \\ \rho_{\text{new}} &= 1 - (1-\rho)^{\ell} - O\left(\frac{\ell^2}{c}\right). \end{aligned}$$

4 Rate amplification for dual-induced SLR

In this section we introduce the notion of smooth locally recoverable sets (SLR) which under non-adaptive and linearity assumptions is shown to be equivalent to LCC. We consider a certain class of SLR, to which we call dual-induced SLR. These are SLR that are obtained by the dual of certain structured sets. The structure of these dual-SLR sets allows us to devise a rate amplification procedures for them. Informally, dual-SLR are sets of tuples (or linear spaces of vectors if the alphabet over which we are working is a field) in which every given entry of a tuple in the set can be recovered using only few queries and in a "smooth" manner, which is to say that the distribution of every query has high min-entropy.

Definition 21 (Author Definition21: Smooth locally recoverable sets (SLR))**.** *Let Σ, P be arbitrary non-empty sets. We say that $C \subseteq \Sigma^P$ is (q, τ, ε)-smooth locally recoverable (SLR for short) if there exists a randomized algorithm* Rec*, called a recovering procedure, that is given as input $p \in P$ and an oracle access to $c \in C$. The recovering procedure outputs either an element of Σ or a symbol $\perp$ which is assumed not to be in Σ. The algorithm* Rec *has the following properties:*

- *For every $(c, p) \in C \times P$, $\text{Rec}^c(p)$ makes at most q queries to c.*
- *For every $c \in C$ and $p, r \in P$ it holds that $\Pr[\text{Rec}^c(p) \text{ queries } c_r] \leq \tau$.*
- *For every $(c, p) \in C \times P$, the random variable $\text{Rec}^c(p) \in \{c_p, \perp\}$, and $\Pr[\text{Rec}^c(p) = \perp] \leq \varepsilon$.*

We assume that for every $p \in P$ whether $\text{Rec}^c(p) = \perp$ is independent of c, and that it is never the case that $\text{Rec}^c(p)$ queries c_p. When Σ is a field and C is a linear subspace of Σ^P, we say that C is linear. In this case, the rate of C is defined as $\dim(C)/|P|$. We will mostly consider non-adaptive SLR. These are SLR in which the joint distribution of queries is independent of c.

We remark that the notion of SLR is very similar to the notion of smooth-codes of [23] for LDC. We now have the following easy claim showing that SLR yield LCC and, assuming linearity, LDC.

Claim 22 (Author Claim22)**.** *Let $C \subseteq \Sigma^P$ be a (q, τ, ε)-SLR. Then, for every $\varepsilon' > 0$, C is a $(q, \delta, \varepsilon + \varepsilon')$-LCC with $\delta = \varepsilon'/(q\tau|P|)$. As a consequence, if C is also linear then C is a $(q, \delta, \varepsilon + \varepsilon')$-LDC.*

Proof. To show that C is an LCC, we devise a local corrector for C. Given an oracle access to $c \in \Sigma^P$, and $p \in P$ as input, the local corrector computes $z = \mathrm{Rec}^c(p)$. If $z = \perp$ then the local corrector returns some arbitrary element of Σ, and otherwise return z. To analyze this local corrector, let $c' \in \Sigma^P$ be such that $\mathrm{dist}(c, c') \leq \delta|P|$. Denote $B = \{p \in P \mid c_p \neq c'_p\}$. Note that conditioned on $\mathrm{Rec}^c(p) \neq \perp$, the local corrector returns c_p successfully if all q queries do not fall into B. The probability that any given query falls into B is bounded above by $\tau|B|$ and so, by the union bound, the probability that some query falls into B is bounded above by $\tau|B|q \leq \varepsilon'$. This proves that C is a $(q, \delta, \varepsilon + \varepsilon')$-LCC. Note that linear LCC are systematic and so every linear LCC induces an LDC. □

In fact, for linear LCCs, the other direction also holds, meaning that such an LCC is an SLR, as we have in the following claim.

Claim 23 (Author Claim23)**.** *Let $C \subseteq \mathbb{F}^P$ be a linear non-adaptive (q, δ, ε)-LCC where $1 - \varepsilon > 1/|\mathbb{F}|$. Then, C is a (q, τ, ε')-SLR with $\tau = q/(\delta|P|)$ and $\varepsilon' = 0$.*

Before we prove the claim, we need to state the following easy to verify fact.

Theorem 24 (Author Fact24)**.** *Let $L \subseteq \mathbb{F}^P$ be a linear subspace, let $p \in P$, let $Q \subseteq P$ and let $x \in \mathbb{F}^{|Q|}$. Then, one of the following cases must hold.*

1. *There is at most one $\alpha \in \mathbb{F}$ for which there exists some $v \in L$ satisfying $v(Q) = x$[2] and $v(p) = \alpha$;*
2. *For every $\alpha \in \mathbb{F}$ there is an equal number of $v \in L$ for which $v(Q) = x$ and $v(p) = \alpha$.*

In particular, either no function (even randomized) of $v(Q)$ can predict $v(p)$ with probability more than $1/|\mathbb{F}|$, when $v \in L$ is randomly chosen uniformly, or $v(Q)$ always determines $v(p)$.

With that, we now prove Claim23[3].

Author proof for Claim23. To show that C is an SLR, we devise a recovering procedure Rec for it, based on the local corrector promised by it being an LCC. Let D[4] be such a local corrector. For every point $p \in P$, we construct a sequence of disjoint sets $Q_p^1, \ldots, Q_p^{m_p} \subseteq P$, where for every i, $c(Q_p^i)$ determines $c(p)$ while satisfying $|Q_p^i| \leq q$, and $m_p \geq \delta|P|/q$. On $p \in P$ and oracle access to $c \in C$, the procedure $\mathrm{Rec}^c(p)$ acts by uniformly choosing $i \in [m_p]$, querying $c(Q_p^i)$, and using it to deduce and output $c(p)$. The correctness of the result of Rec is immediate (since Rec always succeeds, $\varepsilon' = 0$), and indeed the number of queries is no more than q. Since the sets are disjoint, the probability that a point is queried is no more than $\tau = q/(\delta|P|)$. It only remains to show how the assumed sets can be constructed, to conclude that C is a (q, τ, ε')-SLR, which we now turn to do. For every $p \in P$, $Q_p^1, \ldots, Q_p^{m_p}$ are constructed as follows. Set $Q_p^0 = \varnothing$. For $i = 1, 2, \ldots$, set $S_i = Q_p^0 \cup \cdots \cup Q_p^{i-1}$. If $|S_i| > \delta|P|$, halt and set $m_p = i - 1$. Otherwise, it holds that for every $c \in C$, for every

[2] For a set $A = \{a_1, \ldots, a_{|A|}\}$, $v(A)$ denotes the sequence $(v(a_1), \ldots, v(a_{|A|}))$.

[3] This proof is inspired by the proof of [23] of their Theorem1 and by a proof in [41] for a different claim.

[4] We make the slight assumption that $D^c(p)$ never directly queries $c(p)$. If however it does, then similarly C can be shown to be a (q, τ, ε')-SLR for $\tau = 1/((\delta|P|/q) - 1)$.

modification of the coordinates in S_i to some erroneous values, the decoder D correctly outputs $c(p)$ with probability at least $1-\varepsilon$. An equivalent description of this case is the following: for every $c \in C$ and $z : S_i \to \mathbb{F}$, define $c_z \in \mathbb{F}^P$ such that for $r \notin S_i$, $c_z(r) = c(r)$, and for $r \in S_i$, $c_z(r) = z(r)$; the decoder D chooses a set of queries $Q \subseteq P$, $|Q| \leq q$, according to a distribution and applies a function f_Q on $c_z(Q)$; with probability at least $1-\varepsilon$, $f_Q(c_z(Q)) = c(p)$. Since Q is chosen in a manner independent of c and z, one can verify that this implies that there exists some fixed Q for which when $c \in C$ and $z : \mathbb{F} \to S_i$ are chosen randomly in a uniform manner, with probability at least $1-\varepsilon$ (this time over the choice of c and z), $f_Q(c_z(Q)) = c(p)$. Therefore, we can define another function f'_Q that only gets $c(Q \setminus S_i)$, chooses z at random, and outputs $f_Q(c_z(Q))$. If $c \in C$ is chosen uniformly at random, $f'_Q(Q \setminus S_i) = c(p)$ with probability at least $1 - \varepsilon > 1/|\mathbb{F}|$. By Fact24, this implies that $c(Q \setminus S_i)$ determines $c(p)$, for every $c \in C$. We therefore set $Q^i_p = Q \setminus S_i$, and proceed to next i. As this process only halts when $|S_i| > \delta|P|$, and for every i $|S_i| \leq q(i-1)$, we have that indeed $m_p \geq \delta|P|/q$. Further note that by the choice of each Q^i_p, the sets $Q^1_p, \ldots, Q^{m_p}_p$ are disjoint, as required. □

4.1 Dual SLR and their induced SLR

Our construction of SLR sets will be via constructing and analyzing sets which we call dual SLR sets. The SLR will then be induced from these dual SLR. We start by setting some notation. Let P be a non-empty finite set and $\mathbb{F}$ a finite field. We make use of the standard notation $\mathbb{F}^P$ to denote the set of all functions $\{f : P \to \mathbb{F}\}$. Note that $\mathbb{F}^P$ has a natural $\mathbb{F}$-vector space structure where addition is point-wise, namely, for every $f, g \in \mathbb{F}^P$ and $a, b \in \mathbb{F}$ we have that $af + bg \in \mathbb{F}^P$ is defined by $(af + bg)(p) = af(p) + bg(p)$ for all $p \in P$. We consider the natural inner product map $\langle \cdot, \cdot \rangle : \mathbb{F}^P \times \mathbb{F}^P \to \mathbb{F}$ that is defined, for $f, g \in \mathbb{F}^P$, by $\langle f, g \rangle = \sum_{p \in P} f(p)g(p)$. Given $f \in \mathbb{F}^P$, we let $f^\perp = \{g \in \mathbb{F}^P \mid \langle f, g \rangle = 0\}$. Note that $f^\perp$ is a linear subspace of $\mathbb{F}^P$. More generally, given a set $S \subseteq \mathbb{F}^P$ we define the linear subspace $S^\perp = \bigcap_{f \in S} f^\perp$. For $f \in \mathbb{F}^P$ we denote $|f| = |P \setminus f^{-1}(0)|$. For the sake of readability, the field $\mathbb{F}$ and the set P will be omitted from the notation that we are about the introduce in this section. Both will be clear from context. For $p \in P$ define $F_p = \{f \in \mathbb{F}^P \mid f(p) \neq 0\}$. Informally, a dual SLR is a collection of distributions over $\mathbb{F}^P$, one for each point $p \in P$. The distribution D_p, that corresponds to p, outputs a function $g \in \mathbb{F}^P$. We think of g as "passing through" p. We also allow D_p to output a special "failed symbol" $\perp$ with some small probability. A dual SLR has the guarantee that g does not pass through many other points, namely, $|g|$ is bounded, and that the dimension of all functions that can be sampled, when considering all distributions D_p, $p \in P$, is also bounded. Perhaps most importantly is the requirement that for every other fixed $r \in P$, the sampled $g \sim D_p$ is likely to have the property that $g \notin F_r$. Formally,

Definition 25 (Author Definition25: Dual SLR)**.** *Let P be a set, $\mathbb{F}$ a field. Let $D = \{D_p \mid p \in P\}$ be a collection of distributions, where for each $p \in P$, $\mathrm{supp}(D_p) \subseteq F_p \cup \{\perp\}$. Denote $S = \bigcup_{p \in P} \mathrm{supp}(D_p)$. Let L be a linear subspace of $\mathbb{F}^P$ such that $S \subseteq L \cup \{\perp\}$. The pair (D, L) is said to be a $(q, \tau, \varepsilon, \rho)$-dual SLR on $\mathbb{F}^P$ provided the following holds:*

1. *$|g| \leq q$ for all $g \in S \setminus \{\perp\}$.*

2. *For every pair of distinct $p, r \in P$ (not necessarily distinct), it holds that* $\Pr_{g \sim D_p}[g(r) \neq 0 \mid g \neq \perp] \leq \tau$.

3. *For every $p \in P$,* $\Pr[D_p = \perp] \leq \varepsilon$.

4. $\dim(L) \leq (1 - \rho)|P|$.

The linear subspace $L^\perp$ of $\mathbb{F}^P$ is called the induced SLR from the dual SLR (D, L). The parameter τ of a dual-SLR is referred to as its smoothness.

Let (D, L) be a dual SLR. We turn to show that, as the name suggests, the induced SLR $L^\perp$ is indeed an SLR.

Lemma 26 (Author Lemma26)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D, L) be $(q, \tau, \varepsilon, \rho)$-dual SLR on $\mathbb{F}^P$. Then the induced SLR $L^\perp$ is a $(q - 1, \tau, \varepsilon)$-SLR. Furthermore, $L^\perp$ is linear and has rate ρ or larger.*

Proof. The moreover part readily follows since $L^\perp$ is a linear subspace of $\mathbb{F}^P$ and since $\dim(L^\perp) = |P| - \dim(L) \geq \rho|P|$. We describe a recovering procedure for $L^\perp$, namely, a randomized algorithm that is given an oracle access to $f \in L^\perp$ as well as a point $p \in P$ as input. The recovering procedure proceeds as follows:

1. Sample $g \sim D_p$. If $g = \perp$ return $\perp$; Otherwise,
2. Query f on all points $Q = \{r \in P \setminus \{p\} \mid g(r) \neq 0\}$.
3. Return $-\frac{1}{g(p)} \sum_{r \in Q} g(r) f(r)$.

The query complexity of Rec is bounded by $q - 1$ as $|Q| = |g| - 1 \leq q - 1$. The probability that $\perp$ is returned is at most ε by construction. We turn to prove that $\mathrm{Rec}^f(p) \in \{f(p), \perp\}$. By construction, $\mathrm{Rec}^f(p) = \perp$ if and only if $g = \perp$. Assume than that $g \neq \perp$, hence, $g \in \mathrm{supp}(D_p) \subseteq L$. As $f \in L^\perp$ we have that $0 = \langle f, g \rangle$, and so

$$0 = \sum_{r \in P} g(r) f(r) = g(p) f(p) + \sum_{r \in Q} g(r) f(r).$$

As $g \in \mathrm{supp}(D_p) \subseteq F_p$ we have $g(p) \neq 0$, and so

$$f(p) = -\frac{1}{g(p)} \sum_{r \in Q} g(r) f(r) = \mathrm{Rec}^f(p).$$

To conclude the proof, we turn to analyze the smoothness of Rec. First, note that, by construction, f is never queried on p itself. Consider then any $r \in P \setminus \{p\}$. Conditioned on $g \neq \perp$, the function f is queried on r if and only if $g(r) \neq 0$. Thus,

$$\Pr[f(r) \text{ is queried}] = \Pr_{g \sim D_p}[g(r) \neq 0 \mid g \neq \perp] \leq \tau,$$

and the proof follows. □

We now show that the opposite holds as well, that any linear, non-adaptive, SLR induces a dual-SLR.

Lemma 27 (Author Lemma27)**.** *Let P be a set, $\mathbb{F}$ a field, and let $C \subseteq \mathbb{F}^P$ be a linear non-adaptive (q, τ, ε)-SLR with rate ρ. Then for some set D, $(D, C^\perp)$ is a $(q+1, \tau, \varepsilon, \rho)$-dual SLR.*

Proof. Let Rec be a recovering procedure promised by C being a SLR. Assume that Rec uses R random bits. For every point $p \in P$, denote by E_p the set of choices of the random bits $r \in \{0,1\}^R$ for which $\mathrm{Rec}^c(p) = \perp$ (for any $c \in C$). Note that $\mu(E_p \subseteq \{0,1\}^R) \leq \varepsilon$. For any $p \in P$ and $r \in \{0,1\}^R$, $r \notin E_p$, denote by $Q_{p,r} \subseteq P$ the set of query locations which $\mathrm{Rec}(p)$ makes when r is the choice of randomness. Define a function $f_{p,r} : C \to \mathbb{F}$ such that $f(c)$ is the output of $\mathrm{Rec}^c(p)$ when fixing its randomness to r and note that $f_{p,r}(c)$ only depends on $\{c(w) \mid w \in Q_{p,r}\}$, and that $f_{p,r}(c) = c(p)$. Since C is linear, one can easily verify that $f_{p,r}$ is a linear map. Therefore, for some $u_{p,r} \in \mathbb{F}^P$, $f_{p,r}(c) = \langle u_{p,r}, c\rangle$ for every $c \in C$, where $u_{p,r}(w) = 0$ if $w \notin Q_{p,r}$. We have that $c(p) = \langle u_{p,r}, c\rangle$ for every $c \in C$. If we define a function $g_{p,r} \in \mathbb{F}^P$ such that

$$g_{p,r}(w) = \begin{cases} -1, & w = p; \\ u_{p,r}(q), & w \neq p, \end{cases}$$

it follows that $g_{p,r} \in C^\perp$. Note that $|g_{\{p,r\}}| \leq q+1$. For every $p \in P$, define D_p to be the following distribution. To sample from D_p, draw $r \in \{0,1\}^R$ uniformly at random. If $r \in E_p$, output $\perp$; otherwise, output $g_{p,r}$. Set $D = \{D_p \mid p \in P\}$ and $L = C^\perp$. It follows trivially by the definitions that (D, L) is a $(q+1, \tau, \varepsilon, \rho)$-dual SLR. □

4.2 Rate amplification for dual-induced SLR

In this section we describe our first rate amplification procedure for SLR that are induced by dual SLR. Unlike the previous section, it will be more convenient to explicitly state within the notation the set P over which we are working as we will be dealing with several such sets. The field $\mathbb{F}$, however, remains suppressed from the notation as it remains fixed in all SLR under consideration. We start by defining the following map of functions.

Definition 28 (Author Definition28)**.** *Let P be a set and $\mathbb{F}$ a field. For an integer $\ell \geq 1$ we define the map $\Phi : (\mathbb{F}^P)^\ell \to \mathbb{F}^{P^\ell}$ as follows. Let $g_1, \ldots, g_\ell \in \mathbb{F}^P$. The function $\Phi(g_1, \ldots, g_\ell) : P^\ell \to \mathbb{F}$ is defined by*

$$\Phi(g_1, \ldots, g_\ell)(p_1, \ldots, p_\ell) = \prod_{i=1}^{\ell} g_i(p_i)$$

for every $(p_1, \ldots, p_\ell) \in P^\ell$.

Observe that Φ is multi-linear. Further, when $\ell = 2$ and g_1, g_2 are viewed as vectors rather than functions, Φ is the outer product of the vectors.

Definition 29 (Author Definition29)**.** *Let P be a set, $\mathbb{F}$ a field. Let L^P be a linear subspace of $\mathbb{F}^P$. For an integer $\ell \geq 1$, we define*

$$L^{P^\ell} = \mathrm{Span}\{\Phi(g_1, \ldots, g_\ell) \mid g_1, \ldots, g_\ell \in L^P\}.$$

Claim 30 (Author Claim30)**.** *With the notation of Definition29,* $\dim(L^{P^\ell}) \leq \dim(L^P)^\ell$.

Proof. Let $B = \{g_1, \ldots, g_b\}$ be a basis for L^P, where $b = \dim(L^P)$. Define

$$B' = \{\Phi(h_1, \ldots, h_\ell) \mid (h_1, \ldots, h_\ell) \in B^\ell\}.$$

Observe that to prove the claim, it suffices to show that for every $f_1, \ldots, f_\ell \in L^P$ it holds that $\Phi(f_1, \ldots, f_\ell) \in \mathrm{Span}(B')$. As $f_1, \ldots, f_\ell \in L^P$, for every $i \in [\ell]$ we can write $f_i = \sum_{j=1}^{b} \lambda_{i,j} g_j$ with $\lambda_{i,j} \in \mathbb{F}$. We have that

$$\begin{aligned}\Phi(f_1, \ldots, f_\ell) &= \Phi\left(\sum_{j_1=1}^{b} \lambda_{1,j_1} g_{j_1}, \ldots, \sum_{j_\ell=1}^{b} \lambda_{\ell,j_\ell} g_{j_\ell}\right) \\ &= \sum_{j_1,\ldots,j_\ell \in [b]} \left(\prod_{t=1}^{\ell} \lambda_{t,j_t}\right) \cdot \Phi(g_{j_1}, \ldots, g_{j_\ell}),\end{aligned}$$

where the last equality follows by the multi-linearity of Φ. □

Definition 31 (Author Definition31)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be $(q, \tau, \varepsilon, \rho)$-dual SLR. Let $\ell \geq 1$ be an integer. For $p \in P^\ell$ we define the distribution $D_p^{P^\ell}$ as follows. Write $p = (p_1, \ldots, p_\ell)$. To sample an element from $D_p^{P^\ell}$ proceed as follows:*

1. *Sample $g_1 \sim D_{p_1}^P, \ldots, g_\ell \sim D_{p_\ell}^P$ independently.*
2. *If there exists $i \in [\ell]$ such that $g_i = \perp$, return $\perp$; Otherwise*
3. *Return $\Phi(g_1, \ldots, g_\ell)$.*

The collection of distributions $\{D_p^{P^\ell} \mid p \in P^\ell\}$ is denoted by D^{P^ℓ}.

We have the following lemma.

Lemma 32 (Author Lemma32)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be a $(q, \tau, \varepsilon, \rho)$-dual SLR. Let $\ell \geq 1$ be an integer and D^{P^ℓ} as in Definition31. Then, for every $p, r \in P^\ell$,*

$$\Pr_{g \sim D_p^{P^\ell}}[g(r) \neq 0 \mid g \neq \perp] \leq \tau^{\mathrm{dist}(p,r)}.$$

Proof. Write $p = (p_1, \ldots, p_\ell)$, $r = (r_1, \ldots, r_\ell)$. By Definition31, conditioned on $g \neq \perp$ we have that $g = \Phi(g_1, \ldots, g_\ell)$ with $g_i \sim D_{p_i}^P$ for each $i \in [\ell]$ independently. Thus, $g(r) \neq 0$ is the event

$$\Phi(g_1, \ldots, g_\ell)(r_1, \ldots, r_\ell) = \prod_{i=1}^{\ell} g_i(r_i) \neq 0.$$

By the independence of $g_1, \ldots, g_\ell$, and since we are working over a field $\mathbb{F}$ (and so a product is nonzero if and only if each of the terms is nonzero), we get

$$\Pr_{g \sim D_p^{P^\ell}}[g(r) \neq 0 \mid g \neq \perp] = \prod_{i=1}^{\ell} \Pr_{g_i \sim D_{p_i}^P}[g_i(r_i) \neq 0 \mid g_i \neq \perp]. \tag{4.1}$$

Let $T = \{i \in [\ell] \mid p_i \neq r_i\}$. As D^P is a $(q, \tau, \varepsilon, \rho)$-dual SLR, for each $i \in T$ it holds that

$$\Pr_{g_i \sim D^P_{p_i}}[g_i(r_i) \neq 0 \mid g_i \neq \perp] \leq \tau.$$

Substituting to Equation(4.1), we get

$$\Pr_{g \sim D^{P^\ell}_p}[g(r) \neq 0 \mid g \neq \perp] \leq \tau^{|T|},$$

which completes the proof. □

Definition 33 (Author Definition33)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be a $(q, \tau, \varepsilon, \rho)$-dual SLR. For an integer $\ell \geq 1$ let L^{P^ℓ}, D^{P^ℓ} be as in Definition29 and Definition31, respectively. We denote the pair (D^{P^ℓ}, L^{P^ℓ}) by $(D^P, L^P)^\ell$.*

Proposition 34 (Author Proposition34)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be a $(q, \tau, \varepsilon, \rho)$-dual SLR. Then, for every integer $\ell \geq 1$ we have that $(D^P, L^P)^\ell$ is a $(q_\ell, \tau_\ell, \varepsilon_\ell, \rho_\ell)$-dual SLR, where*

$$q_\ell \leq q^\ell, \quad \tau_\ell \leq \tau, \quad \varepsilon_\ell \leq \ell\varepsilon, \quad \rho_\ell \geq 1 - (1-\rho)^\ell.$$

Proof. First note that for every $p \in P^\ell$, the distribution $D^{P^\ell}_p$ is supported on $F^{P^\ell}_p \cup \{\perp\}$. Indeed, if we write $p = (p_1, \ldots, p_\ell)$ then, conditioned on $g \neq \perp$, we have that $g = \Phi(g_1, \ldots, g_\ell)$ where $g_i \in D^P_{p_i}$. Thus,

$$g(p) = \Phi(g_1, \ldots, g_\ell)(p_1, \ldots, p_\ell) = \prod_{i=1}^{\ell} g_i(p_i) \neq 0.$$

Moreover, by Definition29,

$$\bigcup_{p \in P^\ell} \mathrm{supp}(D^{P^\ell}_p) \subseteq L^{P^\ell} \cup \{\perp\}.$$

We turn to show that $q_\ell \leq q^\ell$. Let $p = (p_1, \ldots, p_\ell) \in P^\ell$ and consider any $g \in \mathrm{supp}(D^{P^\ell}_p)$. By Definition31, if $g \neq \perp$ then $g = \Phi(g_1, \ldots, g_\ell)$ where $g_i \in \mathrm{supp}(D^P_{p_i}) \setminus \{\perp\}$. Now, for every $r = (r_1, \ldots, r_\ell) \in P^\ell$ we have that

$$g(r) \neq 0 \quad \Longleftrightarrow \quad \prod_{i=1}^{\ell} g_i(r_i) \neq 0.$$

Since $\mathbb{F}$ is a field, the above is equivalent to $g_i(r_i) \neq 0$ for all $i \in [\ell]$. Hence there are at most q^ℓ points $r \in P^\ell$ for which $g(r) \neq 0$, and so $q_\ell \leq q^\ell$. The bound on the smoothness readily follows by Lemma32. Indeed, consider any pair of distinct $p, r \in \mathbb{F}^{P^\ell}$. We have that $\mathrm{dist}(p, r) \geq 1$ and so, by Lemma32,

$$\Pr_{g \sim D^{P^\ell}_p}[g(r) \neq 0 \mid g \neq \perp] \leq \tau^{\mathrm{dist}(p,r)} \leq \tau. \tag{4.2}$$

To bound the probability that $\perp$ is returned, note that the event $D^{P^\ell} = \perp$ holds only if for some $i \in [\ell]$, $g_i = \perp$. Hence, by the union bound, $\Pr[D^{P^\ell}_p = \perp] \leq \ell\varepsilon$. We conclude the proof by bounding the dimension of L^{P^ℓ}. By assumption, $\dim(L^P) \leq (1-\rho)|P|$. Claim30 then implies that

$$\dim(L^{P^\ell}) \leq \dim(L^P)^\ell \leq ((1-\rho)|P|)^\ell = (1-\rho)^\ell |P^\ell|.$$

□

Discussion on the smoothness $\tau_\ell = \tau$ The downside of the rate amplification procedure that was given in this section is that τ_ℓ does not decrease with ℓ (which is bad as, recall, we wish τ to be small as, by Claim22, the distance δ of the resulted LCC is proportional to $1/\tau$). Indeed, with the notation of Proposition34, $\tau_\ell = \tau$. By examining the proof and Lemma32 one natural idea is to consider an SLR not over the entire set P^ℓ but on some subset of it which is a code with distance, say, $d > 1$. This will indeed guarantee that for every two points p, r we have $\mathrm{dist}(p, r) \geq d$ and so the bound in Equation(4.2) will be τ^d rather than τ. While natural, this idea fails to yield better parameters as the rate-loss incurred by using a code (even an MDS) is larger than the improvement on the rate guaranteed via the rate amplification procedure. In the next sections we give a more elaborate rate amplification procedure (that is based on the one that was given in this section) in which τ does decrease with ℓ. Roughly, $\tau_\ell = (q \cdot \log |P|)^{\mathrm{poly}(\ell)} \tau^\ell$, and so there is a slight loss in the smoothness, which the reader should think as negligible. The query complexity q_ℓ as well as the rate ρ_ℓ and ε_ℓ are all slightly worse than those obtained in the above rate amplification procedure and so the two techniques are incomparable.

4.3 Distance-efficient rate amplification

Let P be a set, and $\mathcal{R}$ a partition of P^2. We denote the part containing p by $[p]_\mathcal{R}$ or $[p]$ when $\mathcal{R}$ is clear from context. We call $(p) = [p] \setminus \{p\}$ the open class of p. For a set $A \subseteq P^2$ we let $(A) = \bigcup_{p \in A} (p)$. Given $p \in P$ we say that $\{p\} \times P \subseteq P^2$ is vertical line and $P \times \{p\}$ is a horizontal line. Horizontal and vertical lines are referred to as axis-parallel lines, and we denote the set of such lines by

$$\mathcal{X} = \bigcup_{p \in P} \{\{p\} \times P, P \times \{p\}\}.$$

For a point $p = (p_1, p_2) \in P^2$ we denote $S_p = (\{p_1\} \times P) \cup (P \times \{p_2\}) \setminus \{p\}$. That is, S_p is the set of points in P^2 of distance exactly 1 from p. Key to our distance-efficient rate amplification procedure is a partition of the "square" P^2 with certain properties.

Definition 35 (Author Definition35: Axis-evasive partitions)**.** *Let P be a set. A partition $\mathcal{R}$ of P^2 is said to be (c, s)-axis evasive if*

1. *For every $p \in P^2$, $|(p)| \leq c$.*
2. *For every $\ell, \ell' \in \mathcal{X}$ (possibly equal), $|\ell' \cap (\ell)| \leq s$.*
3. *For every $p \in P^2$ and $\ell \in \mathcal{X}$, $|[p] \cap \ell| \leq 1$.*

In Section5 we study such partitions. We prove their existence with certain parameters and give explicit constructions. In this section, however, we work with abstract axis-evasive partitions and analyze our rate amplification procedure with respect to the parameters c, s of the axis-evasive partition as well as the number of parts which we typically denote by t.

Claim 36 (Author Claim36)**.** *Let $p, p' \in P^2$ (possibly equal). Then, $|\{r \in S_p \mid (r) \cap S_{p'} \neq \varnothing\}| \leq 4s$.*

Proof. Note that each of $S_p, S_{p'}$ is a subset of the union of two axis-parallel lines. Thus, to prove the claim, it suffices to show that for every $\ell, \ell' \in \mathcal{X}$, not necessarily distinct,

$$|\{r \in \ell \mid (r) \cap \ell' \neq \varnothing\}| \leq s.$$

Let $r_1, \ldots, r_t \in \ell$ be such that $(r_i) \cap \ell' \neq \varnothing$. Note that for every distinct $i, j \in [t]$ it holds that $((r_i) \cap \ell') \cap ((r_j) \cap \ell') = \varnothing$. Indeed, since $\mathcal{R}$ is a partition, if $((r_i) \cap \ell') \cap ((r_j) \cap \ell') \neq \varnothing$ then $r_i \in [r_j]$, but this implies that $|\ell \cap [r_j]| \geq 2$ in contradiction axis evasiveness. Thus,

$$R = \bigcup_{i=1}^{t} ((r_i) \cap \ell')$$

is a disjoint union of size t. However, $R \subseteq (\ell) \cap \ell'$, and so $t \leq |R| \leq |(\ell) \cap \ell'| \leq s$. □

Definition 37 (Author Definition37)**.** *Let P be a set, $\mathbb{F}$ a field. Let $\mathcal{R}$ be a (c, s)-axis evasive partition of P^2. For every $p \in P^2$ define the function $g_{[p]} : P^2 \to \mathbb{F}$ as follows:*

$$g_{[p]}(r) = \begin{cases} 1, & r \in [p]; \\ 0, & \textit{otherwise.} \end{cases}$$

We define $L_{\mathcal{R}} = \{g_{[p]} \mid p \in P^2\}$.

Definition 38 (Author Definition38)**.** *Let P be a set, $\mathbb{F}$ a field. For $S \subseteq P$ define the function $\nu_S : P \to \mathbb{F}$ by*

$$\nu_S(r) = \begin{cases} 0, & r \in S; \\ 1, & \textit{otherwise.} \end{cases}$$

For ease of readability, when S is a singleton $S = \{p\}$, we write ν_p instead of $\nu_{\{p\}}$.

With the notations and definitions above, we are ready to start developing our second rate amplification procedure. We start with the following.

Definition 39 (Author Definition39)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be a $(q, \tau, \varepsilon, \rho)$-dual SLR. Let L^{P^2} be as in Definition29. Let $\mathcal{R}$ be a (c, s)-axis evasive partition of P^2. We define for every $p \in P^2$ the distribution $(D^{P^2}_{\mathcal{R}})_p$ as follows. To sample u from $(D^{P^2}_{\mathcal{R}})_p$:*

1. *Sample $g \sim D^{P^2}_p$.*
2. *If $g = \perp$ return $\perp$; Otherwise, denote $L = \{r \in S_p \mid g(r) \neq 0\}$ and proceed as follows.*
3. *For every $r \in L$ and $w \in (r)$ sample $h_{r,w} \sim D^{P^2}_w$.*
4. *If there exist $r \in L$ and $w \in (r)$ such that either $h_{r,w} = \perp$ or $h_{r,w}(p) \neq 0$ return $\perp$. Otherwise return*

$$u = g\nu_L + \sum_{r \in L} g(r) \sum_{w \in (r)} \frac{h_{r,w}\nu_w}{h_{r,w}(w)}. \tag{4.3}$$

Note that, upon reaching Step(4), u is well-defined as $h_{r,w}(w) \neq 0$ for all $r \in L$ and $w \in (r)$. We denote the collection of distributions $\{(D_{\mathcal{R}}^{P^2})_p \mid p \in P^2\}$ by $D_{\mathcal{R}}^{P^2}$.

We start by analyzing the function u that is given by Equation(4.3) above.

Claim 40 (Author Claim40)**.** *With the notation of Definition39, if $\perp$ is not returned then $u \in F_p$.*

Proof. As $\perp$ was not returned, for every $r \in L$ and $w \in (r)$ it holds that $h_{r,w} \neq \perp$ and $h_{r,w}(p) = 0$. Substituting to Equation(4.3), we get

$$u(p) = g(p)\nu_L(p) = g(p) \neq 0,$$

where the second equality holds as $p \notin L$ and the last inequality follows since $g \in \mathrm{supp}(D_p^{P^2}) \setminus \{\perp\}$. □

Claim 41 (Author Claim41)**.** *With the notation of Definition39, if $\perp$ is not returned then $u \in L^{P^2} + L_{\mathcal{R}}$.*

Proof. Take $f \in (L^{P^2} + L_{\mathcal{R}})^{\perp}$. To prove the claim, it suffices to show that $\langle u, f\rangle = 0$. Indeed, this would imply $u \in ((L^{P^2} + L_{\mathcal{R}})^{\perp})^{\perp} = L^{P^2} + L_{\mathcal{R}}$. As $u \neq \perp$ we have that $g \neq \perp$. Note that

$$\langle g\nu_L, f\rangle = \langle g, f\rangle - \sum_{r\in L} g(r)f(r).$$

Since $g \in \mathrm{supp}(D_p^{P^2})$ we get that $g \in L^{P^2}$. However, $f \in (L^{P^2} + L_{\mathcal{R}})^{\perp} \subseteq (L^{P^2})^{\perp}$, implying $\langle g, f\rangle = 0$. Thus,

$$\langle g\nu_L, f\rangle = -\sum_{r\in L} g(r)f(r). \tag{4.4}$$

Now, fix $r \in L$ and $w \in (r)$. By Definition39, as $u \neq \perp$ we have that $h_{r,w} \neq \perp$ and so $h_{r,w} \in L^{P^2}$. However, by the above, $f \in (L^{P^2})^{\perp}$ and so $\langle h_{r,w}, f\rangle = 0$. Thus,

$$\langle h_{r,w}\nu_w, f\rangle = \langle h_{r,w}, f\rangle - h_{r,w}(w)f(w) = -h_{r,w}(w)f(w).$$

Therefore, for every fixed $r \in L$ one has that

$$\begin{aligned}\left\langle \sum_{w\in(r)} \frac{h_{r,w}\nu_w}{h_{r,w}(w)}, f \right\rangle &= \sum_{w\in(r)} \left\langle \frac{h_{r,w}\nu_w}{h_{r,w}(w)}, f \right\rangle \\ &= \sum_{w\in(r)} \frac{1}{h_{r,w}(w)} \langle h_{r,w}\nu_w, f\rangle \\ &= -\sum_{w\in(r)} f(w). \end{aligned} \tag{4.5}$$

Now, $f \in (L^{P^2} + L_{\mathcal{R}})^{\perp} \subseteq (L_{\mathcal{R}})^{\perp}$ whereas $g_{[r]} \in L_{\mathcal{R}}$, and so

$$0 = \langle f, g_{[r]}\rangle = \sum_{w\in[r]} f(w).$$

Substituting this to Equation(4.5), we get

$$\left\langle \sum_{w\in(r)} \frac{h_{r,w}\nu_w}{h_{r,w}(w)}, f \right\rangle = f(r).$$

Therefore,

$$\begin{aligned}\left\langle \sum_{r\in L} g(r) \sum_{w\in(r)} \frac{h_{r,w}\nu_w}{h_{r,w}(w)}, f \right\rangle &= \sum_{r\in L} g(r) \left\langle \sum_{w\in(r)} \frac{h_{r,w}\nu_w}{h_{r,w}(w)}, f \right\rangle \\ &= \sum_{r\in L} g(r)f(r).\end{aligned}$$

The above equation together with Equation(4.4) yield $\langle u, f\rangle = 0$. □

Claim 42 (Author Claim42)**.** *With the notation of Definition39, for every* $p \in P^2$*,*

$$\Pr[(D_{\mathcal{R}}^{P^2})_p = \perp] \leq 18csq\tau^2 + 2cq\varepsilon.$$

Proof. First, the probability that $g = \perp$ is bounded by ε. Similarly, the probability that for any specific $r \in L$ and $w \in (r)$, $h_{r,w} = \perp$ is bounded by ε. Thus, by the union bound, and since $|L| \leq 2q-1$ and $|(r)| \leq c$, we have that expect with probability $(1+(2q-1)c)\varepsilon \leq 2qc\varepsilon$, the sampling process above will result in $u \neq \perp$. To complete the analysis, we turn to bound the probability that $h_{r,w}(p) = 0$ for some $r \in L$ and $w \in (r)$. Let $L = \{r_1, \ldots, r_{|L|}\}$. While the random variables in L may be dependent, marginally, it holds that for every $i \in [|L|]$ and every fixed $r \in S_p$, $\Pr[r_i = r] \leq \tau$. With this notation, by Definition39, $(D_{\mathcal{R}}^{P^2})_p = \perp$ only if there exist $i \in [|L|]$ and $w \in (r_i)$ such that $h_{r_i,w}(p) \neq 0$. For a fixed $r \in S_p$ define the event E_r in which there exists $w \in (r)$ such that $h_{r,w}(p) \neq 0$, (when conditioned on $h_{r,w} \neq \perp$). Note that this event is with respect to the randomness of sampling $h_r = \{h_{r,w} \mid w \in (r)\}$ whereas r is fixed. By the union bound,

$$\Pr_{h_r}[E_r] \leq \sum_{w\in(r)} \Pr_{h_{r,w}}[h_{r,w}(p) \neq 0 \mid h_{r,w} \neq \perp].$$

Observe first that $w \neq p$. Indeed, as $r \in S_p$, both r and p are on some common axis-parallel line $\ell \in \mathcal{X}$. Thus, $w = p$ would imply $|[r] \cap \ell| \geq 2$ which stands in contradiction to the definition of axis-evasiveness. Consider $w \in (r) \setminus S_p$. As $w \neq p$ we have that $\mathrm{dist}(w, p) = 2$. By Lemma32, as $h_{r,w} \sim D_w^{P^2}$ we have that

$$\Pr_{h_{r,w}}[h_{r,w}(p) \neq 0 \mid h_{r,w} \neq \perp] \leq \tau^2.$$

If, on the other hand, $w \in (r) \cap S_p$ then $\mathrm{dist}(w, p) = 1$, and Lemma32 then implies that

$$\Pr_{h_{r,w}}[h_{r,w}(p) \neq 0 \mid h_{r,w} \neq \perp] \leq \tau.$$

As $|(r)| \leq c$ we conclude that

$$\Pr_{h_r}[E_r] \leq c\tau^2 + \tau|(r) \cap S_p|.$$

Fix $i \in [|L|]$ and consider the random variable r_i. The above equation, together with $|(r_i)| \leq c$, yields

$$\begin{aligned}\Pr_{r_i,h_{r_i}}[E_{r_i}] &\leq \Pr_{r_i,h_{r_i}}[E_{r_i} \mid (r_i) \cap S_p = \varnothing] \\ &\quad + \Pr_{r_i,h_{r_i}}[E_{r_i} \mid (r_i) \cap S_p \neq \varnothing] \Pr_{r_i}[(r_i) \cap S_p \neq \varnothing] \\ &\leq c\tau^2 + (c\tau^2 + c\tau) \Pr_{r_i}[(r_i) \cap S_p \neq \varnothing]. \end{aligned} \tag{4.6}$$

Consider now the set $B = \{r \in S_p \mid (r) \cap S_p \neq \varnothing\}$. As $\mathcal{R}$ is (c, s)-axis evasive, Claim36 implies that $|B| \leq 4s$, and so

$$\Pr_{r_i}[(r_i) \cap S_p \neq \varnothing] = \Pr[r_i \in B] \leq 4s\tau.$$

Substituting to Equation(4.6), we get $\Pr[E_i] \leq 9cs\tau^2$. The proof then follows by taking the union bound over all $i \in [|L|]$ as, indeed, $|L| = 2q - 1$. □

Claim 43 (Author Claim43)**.** *With the notation of Definition39, for every pair of distinct $p, r \in P^2$,*

$$\Pr_{u \sim (D_{\mathcal{R}}^{P^2})_p}[u(r) \neq 0 \mid u \neq \perp] \leq 10csq\tau^2.$$

Proof. By Equation(4.3), u is a linear combination of the (sampled) functions $g\nu_L, \{h_{r,w}\nu_w\}$. To prove the claim, we will show that, with high probability, all these functions evaluate to 0 at the point r, implying $u(r) = 0$. We start by bounding $\Pr[(g\nu_L)(r) \neq 0]$. To this end, consider two cases. First, if $r \in P^2 \setminus S_p$ then, as $L \subseteq S_p$, we have that $\nu_L(r) = 1$ and so in such case

$$\Pr[(g\nu_L)(r) \neq 0] = \Pr[g(r) \neq 0] \leq \tau^2, \tag{4.7}$$

where the last inequality follows by Lemma32 and since $\mathrm{dist}(r, p) = 2$ per our assumption $r \notin S_p$ and since $r \neq p$. If, on the other hand, $r \in S_p$ then, by the definition of L,

$$g(r) \neq 0 \Longrightarrow r \in L \Longrightarrow \nu_L(r) = 0,$$

and so in this case $(g\nu_L)(r) = 0$. Let $L = \{r_1, \ldots, r_{|L|}\}$. Consider a fixed $i \in [|L|]$ and denote $(r_i) = \{w_{i,1}, \ldots, w_{i,b}\}$, where $b \leq c$. Fix $j \in [b]$. We turn to bound $\Pr[(h_{r_i,w_{i,j}}\nu_{w_{i,j}})(r) \neq 0]$. First note that

$$\Pr[(h_{r_i,w_{i,j}}\nu_{w_{i,j}})(r) \neq 0 \mid (r_i) \cap S_r = \varnothing] \leq \tau^2. \tag{4.8}$$

Indeed, conditioned on the event $(r_i) \cap S_r = \varnothing$, either $w_{i,j} = r$ or $\mathrm{dist}(w_{i,j}, r) = 2$. In the first case,

$$(h_{r_i,w_{i,j}}\nu_{w_{i,j}})(r) = h_{r_i,r}(r)\nu_r(r) = 0.$$

In the second case, the bound follows by Lemma32. Second, note that

$$\Pr[(h_{r_i,w_{i,j}}\nu_{w_{i,j}})(r) \neq 0 \mid (r_i) \cap S_r \neq \varnothing] \leq \tau. \tag{4.9}$$

Indeed, as before, we may only consider the case $r \neq w_{i,j}$ and then observe that $\mathrm{dist}(r, w_{i,j}) = 1$ and invoke Lemma32. Now, let $B = \{v \in S_p \mid (v) \cap S_r \neq \varnothing\}$. By

Claim36, and since $\mathcal{R}$ is (c,s)-axis evasive, $|B| \leq 4s$. Recall that $\Pr[r_i = v] \leq \tau$ for every fixed $v \in S_p$, and so

$$\Pr[(r_i) \cap S_r \neq \varnothing] = \Pr[r_i \in B] \leq 4s\tau. \tag{4.10}$$

By Equations(4.8),(4.9),(4.10) we get

$$\Pr[(h_{r_i,w_{i,j}}\nu_{w_{i,j}})(r) \neq 0] \leq \tau^2 + 4s\tau^2 \leq 5s\tau^2.$$

The proof then follows by the union bound over all $i \in [|L|]$ and $j \in [|(w_i)|]$. □

Definition 44 (Author Definition44)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be a $(q,\tau,\varepsilon,\rho)$-dual SLR. Let L^{P^2} be as in Definition29. Let $\mathcal{R}$ be a (c,s)-axis evasive partition of P^2 and let $D^{P^2}_{\mathcal{R}}$ be as in Definition39. We denote by $(D^P, L^P)^2_{\mathcal{R}}$ the pair $(D^{P^2}_{\mathcal{R}}, L^{P^2} + L_{\mathcal{R}})$.*

Proposition 45 (Author Proposition45)**.** *Let P be a set, $\mathbb{F}$ a field, and let (D^P, L^P) be a $(q,\tau,\varepsilon,\rho)$-dual SLR. Let $\mathcal{R}$ be a (c,s)-axis evasive partition of P^2 that consists of t parts. Then, $(D^P, L^P)^2_{\mathcal{R}}$ is a $(q_{\mathcal{R}},\tau_{\mathcal{R}},\varepsilon_{\mathcal{R}},\rho_{\mathcal{R}})$-dual SLR with*

$$\begin{aligned}
q_{\mathcal{R}} &\leq 2cq^3 \\
\tau_{\mathcal{R}} &\leq 10csq\tau^2 \\
\varepsilon_{\mathcal{R}} &\leq 18csq\tau^2 + 2cq\varepsilon \\
\rho_{\mathcal{R}} &\geq 1 - (1-\rho)^2 - \frac{t}{|P|^2}.
\end{aligned}$$

Proof. Claim40 implies that for every $p \in P^2$, $\mathrm{supp}((D^{P^2}_{\mathcal{R}})_p) \subseteq F_p \cup \{\perp\}$. To bound $q_{\mathcal{R}}$, note that by Equation(4.3),

$$|u| \leq |g\nu_L| + \sum_{r \in L} \sum_{w \in (r)} |h_{r,w}\nu_w|.$$

Now, $|g\nu_L| \leq |g| \leq q^2$ and $|h_{r,w}\nu_w| \leq |h_{r,w}| \leq q^2$. Hence, $|u| \leq q^2 + |L|cq^2 \leq 2cq^3$. The stated bounds on $\tau_{\mathcal{R}}$ and $\varepsilon_{\mathcal{R}}$ readily follows by Claim43 and Claim42, respectively. As for the rate, we have that

$$\begin{aligned}
\dim(L^{P^2} + L_{\mathcal{R}}) &\leq \dim(L^{P^2}) + \dim(L_{\mathcal{R}}) \\
&\leq (1-\rho)^2|P|^2 + t,
\end{aligned}$$

where the second inequality follows by Proposition34 and since $\mathcal{R}$ consists of t parts, implying $|L_{\mathcal{R}}| = t$. □

4.4 Proofs of Theorem3 and Corollary4

With the machinery developed in the previous section, and using in a black-box manner, the construction of axis-evasive partitions we obtain in Section5, we are finally ready to prove Theorem3 and Corollary4. We start by giving a more formal statement of Corollary4 (Theorem46, stated at the start of this item).

A remark regarding the error Note that there is another implicit constraint on ρ_{in} and α that originates from the error. Indeed, to make the result non-trivial, one must have $\varepsilon_{\text{out}} < 1$ which, in turn, implies some bound on ℓ and then, through Equation(4.12), a constraint on ρ_{in} and α. However, if that turns out to be a problem for the regime of parameters one is interested in, the probability to output $\perp$ can be reduced by repetition. Thus, by performing an alternating sequence of such error (or failure) reductions and rate amplifications, one can resolve this issue. Note that unlike for LDC, the error reduction has no cost in query complexity, and it certainly has no effect on the smoothness nor on the rate. It does, however, effects the running-time. As mentioned above, our proof relies on an explicit axis-evasive partition that we construct in Section5. Formally,

Theorem 47 (Author Theorem47)**.** *Let P be a set of size q, where q is an odd prime power. Let c be an even integer such that $c+1 \mid q+1$, and $c \leq \sqrt{q}/10$. Then, there exists a $(c, 4c^2)$-axis evasive partition of P^2 with at most $2q^2/(c+1)$ parts.*

Our proof of Theorem46 is done by applying Proposition45 several times, iteratively, where in each iteration we square the size of the set P obtained by the previous iterative step. Note, however, that Theorem47 requires the set size $|P|$ to be an odd prime power q with the property that $c+1 \mid q+1$. It is best to choose c the same in all applications of Proposition45. However, note that if we start an iteration with a set of size q and so end the iteration with a set of size q^2 then the condition will fail to hold at the beginning of the following iteration. Indeed if $c+1 \mid q+1$ then $q \equiv -1 \pmod{c+1}$ and so $q^2 \equiv 1 \pmod{c+1}$. To overcome this technicality, we do not work with the set obtained by the previous iteration as is. Instead, we find a prime–not much smaller than q^2–that has the desired residue -1 modulo $c+1$. To this end we rely on the Siegel–Walfisz Theorem [34, 37] which refines Dirichlet's theorem on primes in arithmetic progressions. The state the Siegel–Walfisz Theorem we set some notation. For an integer $m \geq 1$, we denote Euler's totient function, that counts the positive integers up to m that are relatively prime to m, by $\varphi(m)$. For integers n, m, r, we denote the number of (positive) primes less than or equal to n which are congruent to r modulo m by $\pi(n; m, r)$. The Eulerian logarithmic integral is given by $\text{Li}(x) = \int_2^x dt/\ln t$.

Theorem 48 (Author Theorem48, [34, 37])**.** *For every constant $e \geq 1$ there exists a constant $c = c(e)$ such that the following holds. Let n, m, r be positive integers such that $m \leq (\log n)^e$, and m, r coprimes. Then,*

$$\left|\pi(n; m, r) - \frac{\text{Li}(n)}{\varphi(m)}\right| = O\left(n \cdot 2^{-c\sqrt{\log n}}\right).$$

We have the following straightforward corollary.

Theorem 49 (Author Corollary49)**.** *For every constant $e \geq 1$ there exist constants $c = c(e), n_0 = n_0(e)$ such that the following holds. Let m, r be coprime integers, $m > 0$. Let $n \geq n_0$ be an integer such that $m \leq (\log n)^e$. Then, there exists a prime $p \in [n - \Delta, n]$, where $\Delta = cn/\log n$, such that $p \equiv r \pmod{m}$.*

Proof. To prove the corollary, it suffices to show that $\pi(n;m,r) > \pi(n-\Delta;m,r)$. By Theorem48, there exist constants n_0, c' such that for every $n \geq n_0$,

$$\left|\pi(n;m,r) - \frac{\mathrm{Li}(n)}{\varphi(m)}\right| \leq c' n \cdot 2^{-c\sqrt{\log n}}.$$

Thus, it suffices to show that

$$\frac{\mathrm{Li}(n)}{\varphi(m)} - c' n \cdot 2^{-c\sqrt{\log n}} > \frac{\mathrm{Li}(n-\Delta)}{\varphi(m)} + c'(n-\Delta) \cdot 2^{-c\sqrt{\log(n-\Delta)}}.$$

As we may assume that $\Delta \leq n/2$, it suffices to prove that

$$\mathrm{Li}(n) - \mathrm{Li}(n-\Delta) \geq 2c'\varphi(m) n \cdot 2^{-c\sqrt{\log(n/2)}}. \tag{4.13}$$

It is well-known that

$$\mathrm{Li}(x) = c_1 + \frac{x}{\ln x} + O\left(\frac{x}{\ln^2 x}\right),$$

where $c_1 = \int_{t=0}^{2} dt/\ln t$ is some constant. Therefore,

$$\mathrm{Li}(n) - \mathrm{Li}(n-\Delta) \geq \frac{\Delta}{\ln(n/2)} - \frac{c'' n}{\ln^2 n}.$$

for some constant c''. By our assumption on Δ we can choose the parameter c in the definition of Δ such that the right hand side is bounded below by $n/\ln^2 n$. The proof then follows by Equation(4.13) and noting that $\varphi(m) \leq m \leq (\log n)^e = o(2^{-c\sqrt{\log(n/2)}})$. □

We turn to formally define and analyze the operation of projecting a dual SLR over $\mathbb{F}^P$ on a (large) subset of P.

Definition 50 (Author Definition50)**.** *Let P a set and $P' \subseteq P$. Let $p' \in P'$ and D be a distribution with $\mathrm{supp}(D) \subseteq F_{p'} \cup \{\perp\}$. We define the $D|_{P'}$ as follows: To sample from $D|_{P'}$, sample $f \sim D$. If $f = \perp$, output $\perp$; if $f \in F_{p'}$, output $f|_{P'}$. We refer to $D|_{P'}$ as the distribution D projected to P'.*

Definition 51 (Author Definition51)**.** *Let P be a set, $\mathbb{F}$ a field. Let $D = \{D_p \mid p \in P\}$ be a collection of distributions, where for each $p \in P$, $\mathrm{supp}(D_p) \subseteq F_p \cup \{\perp\}$. Let $P' \subseteq P$. We define $D|_{P'}$ to be the collection D projected to P', that is, $D|_{P'} = \{D_{p'}|_{P'} \mid p' \in P'\}$.*

Definition 52 (Author Definition52)**.** *Let P be a set, $\mathbb{F}$ a field and let L be a linear subspace of $\mathbb{F}^P$. Let $P' \subseteq P$. We denote by $L|_{P'}$ the linear subspace L projected to P', namely, $L|_{P'} = \{f|_{P'} \mid f \in L\}$.*

Claim 53 (Author Claim53)**.** *Let P be a set, $\mathbb{F}$ a field, (D, L) a $(q,\tau,\varepsilon,\rho)$-dual SLR over $\mathbb{F}^P$, and let $P' \subseteq P$. Then, $(D|_{P'}, L_{P'})$ is a $(q,\tau,\varepsilon,\rho')$-dual SLR over $\mathbb{F}^{P'}$, where $\rho' = 1 - \frac{|P|}{|P'|}(1-\rho)$.*

Proof. That the smoothness τ, as well as q and ε, all stay the same after projecting the dual SLR to P', follows immediately from the definitions. The assertion regarding the rate of the induced SLR, ρ', readily follows as we have that

$$\dim(L|_{P'}) \leq \dim(L) \leq (1-\rho)|P| = \left(1 - \left(1 - \frac{|P|}{|P'|}(1-\rho)\right)\right)|P'|.$$

□

Claim 54 (Author Claim54)**.** *There exists a universal constant m_0 such that the following holds. Let P be a set of size $m \geq m_0$. Let $\mathbb{F}$ be a field, and let (D^P, L^P) be a $(q_{\text{in}}, \tau_{\text{in}}, \varepsilon_{\text{in}}, \rho_{\text{in}})$-dual SLR over $\mathbb{F}^P$. Let $c \leq \log m$ be an integer. Then, there exists a set P' of size*

$$|P'| \geq \left(1 - O\left(\frac{1}{\log m}\right)\right) m^2,$$

and a $(q_{\text{out}}, \tau_{\text{out}}, \varepsilon_{\text{out}}, \rho_{\text{out}})$-dual SLR $(D^{P'}, L^{P'})$ over $\mathbb{F}^{P'}$, where

$$\begin{aligned}
q_{\text{out}} &\leq 2cq_{\text{in}}^3, \\
\tau_{\text{out}} &\leq 40c^3 q_{\text{in}} \tau_{\text{in}}^2, \\
\varepsilon_{\text{out}} &\leq 80c^3 q_{\text{in}}(\tau_{\text{in}}^2 + \varepsilon_{\text{in}}), \\
\rho_{\text{out}} &\geq 1 - (1-\rho_{\text{in}})^2 - O(1/c).
\end{aligned}$$

Proof. By Corollary49 applied with n, m, r in the notation of Corollary49 set to $m, c+1, -1$ in the notation of this claim, respectively, there exists some prime $p \leq m$ such that $m - p = O(m/\log m)$, and $c+1 \mid p+1$. Take P' to be an arbitrary subset of P of size p. By Claim53, $(D|_{P'}, L|_{P'})$ is a $(q_{\text{in}}, \tau_{\text{in}}, \varepsilon_{\text{in}}, \rho')$-dual SLR on P', where

$$\rho' = 1 - \frac{m}{p}(1-\rho_{\text{in}}) \geq \rho_{\text{in}} - O\left(\frac{1}{\log m}\right).$$

By Theorem47 applied to P', which observe is indeed applicable as $c+1 \mid p+1$, there exists an explicit $(c, 4c^2)$-axis evasive partition $\mathcal{R}$ of $(P')^2$ with at most $t = 2p^2/(c+1)$ parts. With that partition, we can now apply Proposition45 to $(D|_{P'}, L|_{P'})$ and get that $(D|_{P'}, L|_{P'})^2_{\mathcal{R}}$ is a $(q_{\text{out}}, \tau_{\text{out}}, \varepsilon_{\text{out}}, \rho_{\text{out}})$-dual SLR with the stated parameters. Note that the assertion regarding the rate follows as $c \leq \log m$, □

The following proposition is a more formal and accurate restatement of Theorem3.

Proposition 55 (Author Proposition55)**.** *There exist universal constants $0 < c' < 1$ and $c'', m', \ell' \geq 1$ such that the following holds. Let P be a set of size $m \geq m'$. Let $\mathbb{F}$ be a field, and let (D^P, L^P) be a $(q_{\text{in}}, \tau_{\text{in}}, \varepsilon_{\text{in}}, \rho_{\text{in}})$-dual SLR over $\mathbb{F}^P$. Let $\ell = 2^r$ for an integer $r \geq 1$, and assume that $\ell \geq \ell'$. Let c be an integer such that $c''\ell^2 \leq c \leq c' \log m$. Then, there exists a set P_ℓ of size*

$m^\ell/2 \le |P_\ell| \le m^\ell$, *and a* $(q_\ell, \tau_\ell, \varepsilon_\ell, \rho_\ell)$*-dual SLR* (D^{P_ℓ}, L^{P_ℓ}) *over* $\mathbb{F}^{P_\ell}$, *where*

$$\begin{aligned}
q_\ell &\le (2cq_{\text{in}})^{\ell^{\log 3}},\\
\tau_\ell &= O((c^3 q_{\text{in}})^{\ell^{\log 3}}) \cdot \tau_{\text{in}}^\ell,\\
\varepsilon_\ell &\le O((c^4 q_{\text{in}})^{\ell^{\log 3}}) \cdot (\tau_{\text{in}} + \varepsilon_{\text{in}}),\\
\rho_\ell &\ge 1 - (1-\rho_{\text{in}})^\ell - O\left(\frac{\ell^2}{c}\right),
\end{aligned}$$

where, recall, the log *function is taken base* 2.

Proof. We construct a sequence of $(q_t, \tau_t, \varepsilon_t, \rho_t)$-dual SLR (D^{P_t}, L^{P_t}) for $t = 0, 1, \ldots, r = \log \ell$, and show that the last dual-SLR in the sequence has the stated parameters. The first dual-SLR, (D^{P_0}, L^{P_0}), is taken to be the $(q_{\text{in}}, \tau_{\text{in}}, \varepsilon_{\text{in}}, \rho_{\text{in}})$-dual SLR (D^P, L^P) that is given by the hypothesis of the proposition. After constructing (D^{P_t}, L^{P_t}), we obtain $(D^{P_{t+1}}, L^{P_{t+1}})$ by applying Claim54 to (D^{P_t}, L^{P_t}) with the parameter c taken to be c from the statement of this proposition. Note that, as required by the claim, $c \le \log m$. Note that, by taking m' to be a large enough constant, all other dual SLR in the sequence will have $|P_t| \ge m$ as well, and so we can apply Claim54 to them. Denote $m_t = |P_t|$. We begin by bounding m_t from below. Indeed, by Claim54, and using that $1 - x \ge e^{-2x}$ for $x \le 1/2$, we can pick the constant c'' such that

$$m_t \ge e^{-c''/\log m_{t-1}} m_{t-1}^2 \ge e^{-c''/\log m_0} m_{t-1}^2,$$

where the last inequality follows as, for a large enough constant m', the sequence $(m_t)_t$ is monotone increasing. We invoke Claim82 with $a = e^{c''/(2\log m_0)}$ and $b = 2$ to conclude that

$$m_t \ge m_0^{2^t} e^{-c'' 2^t/\log m_0} \ge \tfrac{1}{2} m_0^{2^t},$$

where the last inequality follows as $t \le r = \log \ell$ and, recall, we take $\ell \le c' \log m$ for a sufficiently small constant $c' > 0$. In particular, $m_r \ge m^\ell/2$ as stated. By Claim54, for every $t \ge 1$ we have $q_t \le 2cq_{t-1}^3$. It is straightforward to prove by that

$$q_t \le (2cq_{\text{in}})^{3^t}, \tag{4.14}$$

which readily implies the assertion regarding the query complexity. We turn to analyze the rate. Denote $\beta_t = 1 - \rho_t$. Claim54 implies that $\beta_t \le \beta_{t-1}^2 + c'''/c$, for some constant $c''' > 0$. By induction on t, we get that $\beta_t \le \beta_0^{2^t} + c'''4^t/c$. Indeed, the base case $t = 0$ is obvious. Now, by the induction hypothesis,

$$\beta_t \le \beta_{t-1}^2 + \frac{c'''}{c} \le \left(\beta_0^{2^{t-1}} + \frac{c'''}{c} 4^{t-1}\right)^2 + \frac{c'''}{c}.$$

One can easily verify that the right hand side is bounded above by the desired bound $\beta_0^{2^t} + c'''4^t/c$ provided that $2^t c'''/c \le 1$. As $t \le r$ and $2^r = \ell$, the latter inequality follows assuming $c'''\ell \le c$. As we assume $c \ge c''\ell^2$, it suffices to choose ℓ' from the statement of the proposition to be a constant larger than the constant c'''/c''. We conclude that,

$$\beta_r \le \beta_0^\ell + O\left(\frac{4^r}{c}\right) = \beta_0^\ell + O\left(\frac{\ell^2}{c}\right),$$

which implies the assertion regarding the rate. As for the smoothness, by Claim54, and using Equation(4.14), we have that

$$\tau_t \leq 40c^3 q_{t-1}\tau_{t-1}^2 \leq 40c^3(2cq_{\text{in}})^{3^{t-1}}\tau_{t-1}^2,$$

from which it is easy to verify that

$$\tau_t \leq (40c^3)^{2^t}(2cq_{\text{in}})^{3^t}\tau_0^{2^t},$$

and the assertion regarding the smoothness readily follows. Last is the error which we leave to the reader to verify. □

We can now easily deduce Theorem46.

Author proof of Theorem46. The proof readily follows from Proposition55 by taking ℓ as defined in Equation(4.12), and with c in the notation of Proposition55 taken to be $c = \Theta(\ell^2/\alpha)$. Note that this choice of parameters satisfies the hypothesis of Proposition55 as indeed implied by Equation(4.11) and by taking c' to be a sufficiently large constant. It is easy to verify that the rate is $1-\alpha$ with our choice of c, ℓ, and the remaining assertions readily follow by Proposition55. □

Claim 82 (Author Claim82, used above; p1:48)**.** *Let $(n_t)_{t\in\mathbb{N}}$ be a sequence of positive integers such that $n_t \geq (n_{t-1}/a)^b$ for some $a, b > 1$. Then, for every $t \geq 1$ we have that $n_t \geq (n_0/a^{h(b,t)})^{b^t}$, where $h(b,t) = \sum_{i=0}^{t-1} 1/b^i$.*

One can prove the following easy claim by induction (author's supplied justification for Claim82).

5 Axis-evasive partitions: 5.2 Explicit constructions

In this section we give explicit constructions of axis-evasive partitions (see Definition35). Our constructions are based on quadratic field extensions. We identify a set P of size q–a prime power–with the finite field $\mathbb{F}_q$ in an arbitrary manner, namely, by using an arbitrary bijection which, for ease of readability, we do not make explicit in the notation. We start by giving some basic background on finite fields. Let $h(x) \in \mathbb{F}_q[x]$ be a degree 2 irreducible monic polynomial. It is a well-known fact that $\mathbb{F}_q[x]/\langle h(x)\rangle$ is a field of size q^2 which we denote, somewhat less informatively, by $\mathbb{F}_{q^2}$. Note that there exists $\alpha \in \mathbb{F}_{q^2}$ such that $h(\alpha) = 0$ (indeed, take $\alpha = x + \langle h(x)\rangle$). Since h is irreducible over $\mathbb{F}_q$ and has degree 2, we can write every element of $\mathbb{F}_{q^2}$ in the form $a + \alpha b$, where $a, b \in \mathbb{F}_q$, in a unique manner. That is, we can identify in the set-theoretic level, $\mathbb{F}_{q^2}$ with $\mathbb{F}_q + \alpha\mathbb{F}_q$. Using this identification, we identify P^2 with $\mathbb{F}_{q^2}$ in the natural way, namely, a point $(a, b) \in P^2$ is identified with $a + \alpha b$ in $\mathbb{F}_{q^2}$. Note that, with this identification, the horizontal lines in P^2 are of the form $b\alpha + \mathbb{F}_q$ where $b \in \mathbb{F}_q$ can be thought of as the fixed height of the line. Similarly, the vertical lines are given by $b + \alpha\mathbb{F}_q$. Given $\delta \in \mathbb{F}_{q^2} \setminus \{0\}$, we say that $\ell_\delta = \delta\mathbb{F}_q \subseteq \mathbb{F}_{q^2}$ is the line through the origin with slope δ. Our construction of exis-evasive partitions is based on an equivalence relation that we are about

to define. The partition is then obtained by considering the respective equivalence classes. We begin the construction by ignoring the "origin" $0 \in \mathbb{F}_{q^2}$ and work only with $\mathbb{F}_{q^2} \setminus \{0\}$. Note that this is the set of invertible elements of $\mathbb{F}_{q^2}$ which has a group structure under the field multiplication. When referring to this multiplicative group we write $(\mathbb{F}_{q^2})^\times$. Let $\beta \in (\mathbb{F}_{q^2})^\times$. Denote by $o(\beta)$ the order of β in the multiplicative group $(\mathbb{F}_{q^2})^\times$. It will be convenient to denote $c = o(\beta) - 1$. We define an equivalence relation on $(\mathbb{F}_{q^2})^\times$, parameterized by β, as follows: For $\gamma, \delta \in (\mathbb{F}_{q^2})^\times$

$$\gamma \sim \delta \quad \iff \quad \gamma\delta^{-1} \in \langle\beta\rangle, \tag{5.1}$$

where $\langle\beta\rangle$ is the subgroup of $(\mathbb{F}_{q^2})^\times$ that is generated by β. Observe that this is an equivalence relation. Indeed, the classes are the different cosets, that is, the elements of the quotient group $(\mathbb{F}_{q^2})^\times/\langle\beta\rangle$. For completeness, we quickly prove that this is an equivalence relation: as $1 \in \langle\beta\rangle$, we have that $\gamma \sim \gamma$. Secondly, if $\gamma\delta^{-1} \in \langle\beta\rangle$ then $\delta\gamma^{-1} \in \langle\beta^{-1}\rangle = \langle\beta\rangle$ which establishes symmetry. As for transitivity, if $\gamma \sim \delta$ and $\delta \sim \varepsilon$ then

$$\gamma\varepsilon^{-1} = \gamma(\delta^{-1}\delta)\varepsilon^{-1} = (\gamma\delta^{-1})(\delta\varepsilon^{-1}) \in \langle\beta\rangle.$$

One can easily see that the equivalence class of an element $\gamma \in (\mathbb{F}_{q^2})^\times$ is $[\gamma] = \gamma\langle\beta\rangle = \{\gamma, \beta\gamma, \ldots, \beta^c\gamma\}$. Note further that $|[\gamma]| = c + 1$. Indeed, if there are $0 \le j < i \le c$ such that $\beta^i\gamma = \beta^j\gamma$ then $0 = (\beta^i - \beta^j)\gamma = (\beta^{i-j} - 1)\beta^j\gamma$, which is a contradiction as none of the factors in the product is zero. In the following claim we show that, under some conditions on α, β, the second property of axis-evasiveness is met by the construction above. We mention already here that the third condition in Definition35 is not met by the construction as is (regardless of the choice of α, β), and we will alter it afterwards to meet that property as well.

Claim 57 (Author Claim57)**.** *Assume that $\langle\beta\rangle \cap \ell_\alpha = \langle\beta\rangle \cap \ell_{\alpha^{-1}} = \varnothing$ and that $\langle\beta\rangle \cap \mathbb{F}_q = \{1\}$. Then, for every $\ell, \ell' \in \mathcal{X}$ (not necessarily distinct) it holds that $\ell' \cap (\ell) \le c$.*

Proof. Recall that $(\gamma) = \{\beta\gamma, \ldots, \beta^c\gamma\}$. Thus,

$$\bigcup_{\gamma\in\ell}(\gamma) = \bigcup_{\gamma\in\ell}\bigcup_{i=1}^{c}\{\beta^i\gamma\} = \bigcup_{i=1}^{c}\beta^i\ell.$$

Therefore,

$$\ell' \cap (\ell) = \ell' \cap \bigcup_{\gamma\in\ell}(\gamma) = \bigcup_{i=1}^{c}(\ell' \cap \beta^i\ell). \tag{5.2}$$

Fix $i \in [c]$ and consider two cases. First, if ℓ is vertical, namely, $\ell = b + \alpha\mathbb{F}_q$ for some $b \in \mathbb{F}_q$, then $\beta^i\ell = \beta^i b + \alpha\beta^i\mathbb{F}_q$. Since, by assumption, $\langle\beta\rangle \cap \mathbb{F}_q = \{1\}$ we have that $\alpha\beta^i\mathbb{F}_q \neq \alpha\mathbb{F}_q$ and so the line $\beta^i\ell$ is not vertical. As, by assumption, $\langle\beta\rangle \cap \ell_{\alpha^{-1}} = \varnothing$, we have that $\alpha\beta^i \notin \mathbb{F}_q$ and so the line $\beta^i\ell$ is not horizontal either. Second, consider the case that ℓ is horizontal $\ell = b\alpha + \mathbb{F}_q$ for some $b \in \mathbb{F}_q$. Then, $\beta^i\ell = b\alpha\beta^i + \beta^i\mathbb{F}_q$. Per our assumption that $\langle\beta\rangle \cap \ell_\alpha = \varnothing$, we have that $\beta^i\mathbb{F}_q \neq \alpha\mathbb{F}_q$ and so the line $\beta^i\ell$ is not vertical. As we assume $\langle\beta\rangle \cap \mathbb{F}_q = \{1\}$, we have that $\beta^i\mathbb{F}_q \neq \mathbb{F}_q$, and so the line $\beta^i\ell$ cannot be horizontal either. To summarize, we have that $\beta^i\ell \notin \mathcal{X}$. However, $\ell' \in \mathcal{X}$ and so $\beta^i\ell$ and ℓ' are two distinct lines. As such, the two lines intersect in at most one point. Equation(5.2) then yield $|\ell' \cap (\ell)| \le c$. □

Informal discussion regarding the third property As mentioned above, the partition of $(\mathbb{F}_{q^2})^\times$ as defined above does not have the third property required for axis-evasiveness. Namely, there are $\gamma \in (\mathbb{F}_{q^2})^\times$ such that $[\gamma]$ intersects some axis-parallel line at more than one point. To get some idea on which equivalence classes $[\gamma]$ are problematic, let us first ask when do $\gamma, \beta\gamma$ are on some common axis-parallel line. We first observe that two points $\delta, \varepsilon \in (\mathbb{F}_{q^2})^\times$ are on a common axis-parallel line if and only if $\delta - \varepsilon \in \{1, \alpha\}\mathbb{F}_q$. Thus, γ and $\beta\gamma$ are on the same axis-parallel line if and only if $\gamma - \beta\gamma = (1-\beta)\gamma \in \{1, \alpha\}\mathbb{F}_q$. This is equivalent to saying that γ is on one of the two lines through the origin with slopes $1/(1-\beta), \alpha/(1-\beta)$. More generally, $[\gamma]$ intersects with some axis-parallel line in more than one point if and only if $\beta^i\gamma - \beta^j\gamma \in \{1, \alpha\}\mathbb{F}_q$ for some $0 \leq j < i \leq c$. Equivalently, γ is on a line ℓ_δ with

$$\delta \in \left\{ \frac{1}{\beta^i - \beta^j}, \frac{\alpha}{\beta^i - \beta^j} \;\middle|\; 0 \leq j < i \leq c \right\}. \tag{5.3}$$

The key observation is that although there are a fair amount of "bad" points γ, they are all contained in a small number of lines. By "small" here we mean that the number is polynomial in c and is independent of q. Thus, the hope is that by redefining the partition on these few problematic lines we will not harm the previous analysis by much. Indeed, no matter how we alter the partition restricted to these lines, if we make sure none of them is axis-parallel (by requiring more properties from α, β) then each of these lines intersect an axis-parallel line at one point. As a result, the bound obtained in Claim57 will deteriorate proportionally to the number of lines above. The only small technical issue is that even if $\gamma \in \ell_\delta$ for some slope δ as above, it is not the case that $[\gamma] \subseteq \bigcup_\varepsilon \ell_\varepsilon$ where ε is taken from the set of slopes given by Equation(5.3). As we wish to alter the partition defined above, it would be cleaner to have all of the points in $[\gamma]$ of a problematic point γ contained in the set of points on which we redefine the partition. Thus, we "close" the set of slopes given by Equation(5.3) to multiplication by β. Ending the informal discussion and returning to the formal analysis, we consider the set of slopes.

$$\Delta = \left\{ \frac{\beta^k}{\beta^i - \beta^j}, \frac{\alpha\beta^k}{\beta^i - \beta^j} \;\middle|\; 0 \leq j < i \leq c \text{ and } 0 \leq k \leq c \right\}. \tag{5.4}$$

Further define the set of all points in $(\mathbb{F}_{q^2})^\times$ covered by the lines with slopes from Δ by $U = \bigcup_{\delta \in \Delta} \ell_\delta$. This definition of Δ indeed fixes the technical caveat discussed above, as the following claim states.

Claim 58 (Author Claim58)**.** *For every $\gamma \in (\mathbb{F}_{q^2})^\times$ either $[\gamma] \subseteq U$ or $[\gamma] \cap U = \varnothing$.*

Proof. If an element $\varepsilon \in U$ then $\varepsilon \in \ell_\delta$ for some $\delta \in \Delta$. Note that $\beta\varepsilon \in \ell_{\beta\delta}$ and that $\beta\delta \in \Delta$. Hence, $\beta\varepsilon \in U$. Therefore, $\varepsilon \in U \implies \varepsilon\langle\beta\rangle \subseteq U$. Assume now that $[\gamma] \cap U \neq \varnothing$, and take $\gamma\beta^i \in U$. By the above, $\gamma\beta^i\langle\beta\rangle \subseteq U$. The proof then follows as $\gamma\beta^i\langle\beta\rangle = \gamma\langle\beta\rangle = [\gamma]$. □

Define a new partition of $\mathbb{F}_{q^2}$ (including 0) which agrees with the one that is given by Equation(5.1) on $\mathbb{F}_{q^2}^\times \setminus U$. By Claim58, this is well-defined. The new partition, restricted to U, is done as follows. Let $\delta_0 \in \Delta$ be an arbitrary

element. Note that

$$U = \ell_{\delta_0} \cup \bigcup_{\delta \in \Delta \setminus \{\delta_0\}} (\ell_\delta \setminus \{0\})$$

is a disjoint union. To partition U, we partition ℓ_{δ_0} as well as each of $\ell_\delta \setminus \{0\}$ where $\delta \in \Delta \setminus \{\delta_0\}$ in an arbitrary way provided it has the least number of parts under the conditions that each part has size at most $c+1$. For ease of readability, we denote by $[\gamma]$ the class with respect to the new partition.

Claim 59 (Author Claim59)**.** *Assume, on top of the assumptions of Claim57 that for every $\delta \in \Delta$, $\ell_\delta \notin \mathcal{X}$. Then, the new partition defined above is $(c, 4c^2)$-axis evasive.*

Proof. First, observe that by construction, every class intersects any axis-parallel line in at most one point. Indeed, classes that are outside of U have this property by the definition of U as can be easily verified (and discussed above). Moreover, by the way we redefined the partition restricted to U, every class that is a subset of U is also a subset of a line ℓ_δ for some $\delta \in \Delta$. As $\ell_\delta \notin \mathcal{X}$ by hypothesis, we have that the line and, as a result, the class it contains, intersects any axis-parallel line in at most one point. This establishes the third property of axis-evasiveness. The second property follows as, by construction, every part has size at most $c+1$. Moving on to the second property, consider $\ell, \ell' \in \mathcal{X}$, not necessarily distinct. As outside of U the partition is defined as before, Claim58 yields

$$\left| \ell' \cap \bigcup_{\gamma \in \ell \setminus U} (\gamma) \right| \leq c. \tag{5.5}$$

Take $\gamma \in U \cap \ell$. Since, by construction $(\gamma) \subseteq \ell_\delta$ for some $\delta \in \Delta$, and since by hypothesis $\ell_\delta \notin \mathcal{X}$ we have that $|\ell' \cap \ell_\delta| = 1$ and $(\gamma) \cap \ell' \subseteq \ell_\delta \cap \ell'$. Therefore, $|(\gamma) \cap \ell'| \leq 1$. Together with Equation(5.5) we get that $|\ell' \cap (\ell)| \leq c + |U \cap \ell|$. Now, since $\ell \in \mathcal{X}$ and every line ℓ_δ with slope $\delta \in \Delta$ is not in $\mathcal{X}$ we have that $|\ell \cap \ell_\delta| = 1$. Thus, $|U \cap \ell| \leq |\Delta|$ which implies $|\ell' \cap (\ell)| \leq c + |\Delta|$. To conclude the proof, we turn to bound $|\Delta|$. It is straightforward to give a bound of $O(c^3)$ though one can optimize the bound a bit. Indeed, with the notation of Equation(5.4), by multiplying by $\beta^{-\min(j,k)}$, one can rewrite

$$\begin{aligned} \Delta = &\left\{ \frac{1}{\beta^i - \beta^j}, \frac{\alpha}{\beta^i - \beta^j} \;\middle|\; 0 < j < i \leq c \right\} \\ &\cup \left\{ \frac{\beta^j}{\beta^i - 1}, \frac{\alpha\beta^j}{\beta^i - 1} \;\middle|\; 0 < i \leq c,\ 0 \leq j \leq c \right\}. \end{aligned} \tag{5.6}$$

Thus, $|\Delta| \leq 3c^2$, and the proof follows. □

We summarize the discussion so far.

Proposition 60 (Author Proposition60)**.** *Let $\mathbb{F}_q$ be finite field. Let $h(x) \in \mathbb{F}_q[x]$ be a degree 2 irreducible monic polynomial, and consider the field $\mathbb{F}_q[x]/\langle h(x) \rangle$ which we denote by $\mathbb{F}_{q^2}$. Let $\alpha, \beta \in \mathbb{F}_{q^2}$ be two elements satisfying:*

1. *$h(\alpha) = 0$,*
2. *$\langle \beta \rangle \cap \mathbb{F}_q = \{1\}$,*

3. $c+1 = o(\beta) \leq \sqrt{q}/10$,

4. $\langle\beta\rangle \cap \ell_\alpha = \langle\beta\rangle \cap \ell_{\alpha^{-1}} = \varnothing$,

5. $(\langle\beta\rangle - \langle\beta\rangle) \cap \mathbb{F}_q = \{0\}$,

6. $(\langle\beta\rangle - \langle\beta\rangle) \cap \ell_\alpha = (\langle\beta\rangle - \langle\beta\rangle) \cap \ell_{\alpha^{-1}} = \{0\}$.

Then, there exists a partition of $(\mathbb{F}_q)^2$ that is $(c, 4c^2)$-axis-evasive, where $c = o(\beta) - 1$. The number of parts in the partition is bounded above by $2q^2/(c+1)$.

To prove Proposition60 we need the following easy claim.

Claim 61 (Author Claim61)**.** *Let $\delta \in (\mathbb{F}_{q^2})^\times$ be such that $\delta \notin \mathbb{F}_q \cup \ell_\alpha$ then, $\ell_\delta \notin \mathcal{X}$.*

Proof. Write $\delta = a + \alpha b$ with $a, b \in \mathbb{F}_q$. Then, $\ell_\delta = (a + \alpha b)\mathbb{F}_q$. Observe that if ℓ_δ is vertical then $a = 0$ and so $\delta \in \ell_\alpha$. Similarly, if ℓ_δ is horizontal then $b = 0$ implying $\delta \in \mathbb{F}_q$. □

Author proof of Proposition60. To bound the number of parts, recall that in the original partition, each part has size $c+1$. Moreover, in the altered partition we partition each line ℓ_δ with slope $\delta \in \Delta$ (excluding the origin from all but for one of the lines ℓ_{δ_0}) to parts of size $c+1$ each, except for possibly one part. As $|\Delta| \leq 3c^2$, the number of parts it bounded by

$$\frac{q^2-1}{c+1} + |\Delta|\left(1 + \frac{q}{c+1}\right) \leq \frac{q^2-1}{c+1} + 6cq \leq \frac{2q^2}{c},$$

where the last inequality follows by our assumption that $o(\beta) \leq \sqrt{q}/10$. To conclude the proof of the proposition, it suffices to show that for every $\delta \in \Delta$ it holds that $\ell_\delta \notin \mathcal{X}$. By Claim61, it suffices to prove that $\delta \notin \mathbb{F}_q \cup \ell_\alpha = \{1, \alpha\}\mathbb{F}_q$. There are two types of slopes $\delta \in \Delta$, according to whether they appear in the first or second set in Equation(5.6). The first kind is of the form

$$\delta = \frac{\alpha^k}{\beta^i - \beta^j},$$

with $0 < j < i \leq c$ and $k \in \{0, 1\}$. If $\delta \in \{1, \alpha\}\mathbb{F}_q$ then $\delta^{-1} \in \{1, \alpha^{-1}\}\mathbb{F}_q$ and so $\beta^i - \beta^j \in \{\alpha^k, \alpha^{k-1}\}\mathbb{F}_q$ in contradiction to our hypothesis. Consider now the other kind of slope

$$\delta = \frac{\alpha^k \beta^j}{\beta^i - 1}$$

where $0 < i \leq c$, $0 \leq j \leq c$ and $k \in \{0, 1\}$. If $\delta \in \{1, \alpha\}\mathbb{F}_q$ then $\delta^{-1} \in \{1, \alpha^{-1}\}\mathbb{F}_q$ and so $(\beta^i - 1)\beta^{-j} \in \{\alpha^k, \alpha^{k-1}\}\mathbb{F}_q$. Note that $(\beta^i - 1)\beta^{-j} = \beta^{i-j} - \beta^{-j} \in \langle\beta\rangle - \langle\beta\rangle$ and so we again get a contradiction. □

We are now ready to prove Theorem47. For the sake of readability, we repeat its statement here.

Theorem 62 (Author Theorem62)**.** *Let P be a set of size q, where q is an odd prime power. Let c be an even integer such that $c+1 \mid q+1$, and $c \leq \sqrt{q}/10$. Then, there exists a $(c, 4c^2)$-axis evasive partition of P^2 with at most $2q^2/(c+1)$ parts.*

Proof. As above, we identify P^2 with $\mathbb{F}_{q^2}$. It is a well-known fact that the multiplicative group $(\mathbb{F}_{q^2})^\times$ is cyclic. A basic result in group theory states that a cyclic group has a (unique) subgroup of every given size which divides the group size. Now, $|(\mathbb{F}_{q^2})^\times| = q^2 - 1 = (q-1)(q+1)$. Thus, as $c+1 \mid q+1$, there exists a subgroup H of $(\mathbb{F}_{q^2})^\times$ of size $c+1$. The subgroup H is cyclic, being a subgroup of a cyclic group. Let β be a generator for H. We first prove that β satisfies those hypothesis of Proposition60 that do not involve α, namely, conditions(2) and(5).

Claim 63 (Author Claim63)**.** $(\langle\beta\rangle - \langle\beta\rangle) \cap \mathbb{F}_q = \{0\}$ *and* $\langle\beta\rangle \cap \mathbb{F}_q = \{1\}$.

Proof. Assume towards a contradiction that $\beta^i - \beta^j \in \mathbb{F}_q$ for some $0 \le j < i \le c$. Since $x^q = x$ for every $x \in \mathbb{F}_q$, we get

$$\beta^i - \beta^j = (\beta^i - \beta^j)^q = \beta^{iq} - \beta^{jq},$$

where the last equality follows since q is divisible by the characteristic of the field. Recall that $o(\beta) = c+1 \mid q+1$ and so $\beta^{i(q+1)} = 1$, implying $\beta^{iq} = \beta^{-i}$. Thus,

$$\beta^i - \beta^j = \frac{1}{\beta^i} - \frac{1}{\beta^j} = \frac{\beta^j - \beta^i}{\beta^{i+j}}.$$

As $\beta^i \neq \beta^j$ the above equation implies $\beta^{i+j} = -1$, and so $-1 \in H$. Since q is odd, the characteristic of the field $\mathbb{F}_{q^2}$ is odd and so $o(-1) = 2$. Lagrange's Theorem then implies that $2 \mid |H| = c+1$, which stands in contradiction to c being even. To prove that $\langle\beta\rangle \cap \mathbb{F}_q = \{1\}$, take β^i with $0 < i \le c$. If $\beta^i \in \mathbb{F}_q$ then $\beta^{iq} = \beta^i$. On the other hand, we proved above that $\beta^{iq} = \beta^{-i}$, and so $\beta^i = \beta^{-i}$ implying $\beta^{2i} = 1$. Therefore, $o(\beta) = c+1 \mid 2i$, but this is impossible as $0 < i \le c$ and, recall, c is even. □

We proceed with the proof of Theorem47 by finding $\alpha \in \mathbb{F}_{q^2}$ that, together with the already chosen β, satisfies the remaining conditions in the hypothesis of Proposition60. Since $\mathbb{F}_{q^2}$ is a quadratic field extension of $\mathbb{F}_q$, every element $\gamma \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ has degree 2. That is, the minimal polynomial h_γ of every such γ over $\mathbb{F}_q$ is of degree 2 (and can be made monic by dividing by the leading coefficient, if necessary). Indeed, $\deg(h_\gamma)$ cannot equal 1 as this would imply $\gamma \in \mathbb{F}_q$. On the other hand,

$$2 = [\mathbb{F}_{q^2} : \mathbb{F}_q] = [\mathbb{F}_{q^2} : \mathbb{F}_q(\gamma)][\mathbb{F}_q(\gamma) : \mathbb{F}_q] = [\mathbb{F}_{q^2} : \mathbb{F}_q(\gamma)] \deg(h_\gamma),$$

which shows that if $\deg(h_\gamma) \neq 1$ then $\deg(h_\gamma) = 2$. Thus, condition(1) in the hypothesis of Proposition60 holds for every element in $\mathbb{F}_{q^2} \setminus \mathbb{F}_q$. Hence, to prove that all the remaining conditions in the hypothesis of Proposition60 hold, it suffices to prove that there exists $\alpha \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ which satisfies conditions(4) and(6). To this end, pick a set of "slopes" $\Delta' = \{\delta_1, \ldots, \delta_{q+1}\} \subseteq (\mathbb{F}_{q^2})^\times$ such that $(\mathbb{F}_{q^2})^\times$ is the disjoint union

$$(\mathbb{F}_{q^2})^\times = \bigcup_{\delta \in \Delta'} (\ell_\delta \setminus \{0\}).$$

For example, $\Delta' = \{a + \alpha \mid a \in \mathbb{F}_q\} \cup \{1\}$ will do. For $\delta \in (\mathbb{F}_{q^2})^\times$ let

$$I_\delta = |\langle\beta\rangle \cap \ell_\delta| + |(\langle\beta\rangle - \langle\beta\rangle) \cap (\ell_\delta \setminus \{0\})|.$$

Since the $\ell_\delta \setminus \{0\}$ with $\delta \in \Delta'$ are disjoint, $0 \notin \langle\beta\rangle$, and since $|\langle\beta\rangle| = c+1$ and $|\langle\beta\rangle - \langle\beta\rangle| \leq (c+1)^2$, we have that

$$\mathbb{E}_\delta[I_\delta] \leq \frac{(c+1)^2 + (c+1)}{q+1} \leq \frac{2(c+1)^2}{q+1},$$

where δ is sampled uniformly from Δ'. By Markov's inequality, for at least 3/4 of the elements $\delta \in \Delta'$ it holds that

$$|I_\delta| \leq \frac{8(c+1)^2}{q+1}.$$

Note that $(\mathbb{F}_{q^2})^\times$ is also a disjoint union of $\{\ell_{\delta^{-1}} \setminus \{0\} \mid \delta \in \Delta\}$. Thus, using the same argument as above, we get that for at least 1/2 the elements $\delta \in \Delta'$, both $|I_\delta|$ and $|I_{\delta^{-1}}|$ are bounded by $8(c+1)^2/(q+1)$. But, as $c \leq \sqrt{q}/10$, this bound is strictly smaller than 1, implying that $I_i = I_{q+1-i} = 0$. That is, at least half the elements $\delta \in \Delta'$ satisfy conditions(4) and(6). Take α to be any of these elements. To conclude, we found α and β for which all the conditions in the hypothesis of Proposition60 are met, and the proof follows. □

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